Yingshi Huang

CONTACT INFORMATION

California, United States
+1 310 873 7956
yingshi@ucla.edu
https://quantithinker-yingshi.github.io/

EDUCATION

University of California, Los Angeles (UCLA), California, United States
Ph.D. in Education — Sep. 2023 - Present
M.S. in Statistics & Data Science — Sep. 2024 - Present

- Research Interest: Latent space model; Network analysis; Casual inference
- Advisor: Prof. Minjeong Jeon

Beijing Normal University, Beijing, China
M.A. in Psychology — Sep. 2020 – Jun. 2023

South China Normal University, Guangzhou, China
B.S. in Psychology — Sep. 2016 – Jun. 2020

PUBLICATIONS

(* correspondent author, + co-first author)

Educational and Psychological Measurement

1. **Huang, Y.**, Wang, S., Pan, Y., Lu, X., & Chen, P.* (under review). A motivation-based cognitive diagnostic model for disengaged responses detection.
2. Sun, X., **Huang, Y.**, & Song, N.* (under review) Estimating the Q-matrix for cognitive diagnostic models: Based on partial known q-entries.
3. Yuan, L., **Huang, Y.**, & Chen, P.* (2025). Online calibration for multidimensional CAT with polytomously scored items: A neural network–based approach. *Journal of Educational and Behavioral Statistics*, *0*(0). doi:10.3102/10769986251315531.
4. Chen, P.*, Dai, Y., **Huang, Y.** (2023). Test mode effect: Sources, detection, and applications (in Chinese). *Advances in Psychological Science*, *31*(10), 1966–1980. doi: 10.3724/SP.J.1042.2023.01966
5. Yuan, L., **Huang, Y.**, Li, S., & Chen, P.* (2023). Online calibration in multidimensional computerized adaptive testing with polytomously scored items. *Journal of Educational Measurement*, *60*(3), 476-500. doi: 10.1111/jedm.12353.
6. **Huang, Y.**, Ren, H., & Chen, P.* (2023). Item selection methods with exposure and time control for computerized classification test. *British Journal of Mathematical and Statistical Psychology*, *76*, 52–68. doi: 10.1111/bmsp.12281.
7. Ren, H., **Huang, Y.**, & Chen, P.* (2022). Types, characteristics and application of termination rules in computerized classification testing (in Chinese). *Advances in Psychological Science*, *30*(5), 1168. doi: 10.3724/SP.J.1042.2022.01168.

Substantive Topics

1. **Huang, Y.**, Luo, J., Shetty, V., & Jeon, M.* (under review). The power of social talk: A longitudinal network analysis of conversations in fostering interdisciplinary collaboration.
2. Ren, J., Luo, J., **Huang, Y.**, Shetty, V., & Jeon, M.* (2025). Mapping the mHealth nexus: A semantic analysis of mHealth scholars' research propensities following an interdisciplinary training institute. *Applied Sciences*, *15*(11), 6252. doi: 10.3390/app15116252.
3. Ni, Y.*, **Huang, Y.**, Zhao, S., Ma, Y., & Fang, M. (2025). Development and evaluation of the shortened Chinese version of school social behavior scales-2 using item response theory and psychometric network analysis. *Measurement and Evaluation in Counseling and Development*, 1-19. doi: 10.1080/07481756.2025.2499965.

4. Lu, A.*+, Deng, R.+, **Huang, Y.+**, Song, T., Shen, Y., Fan, Z., & Zhang, J. (2022). The roles of mobile app perceived usefulness and perceived ease of use in app-based Chinese and English learning flow and satisfaction. *Education and Information Technologies*, 1-22. doi: 10.1007/s10639-022-11036-1.
5. Ni, Y., Tein, J. Y., Zhang, M.*, Zhen, F., Huang, F., **Huang, Y.**, Yao, Y., & Mei, J. (2020). The need to belong: A parallel process latent growth curve model of late life negative affect and cognitive function. *Archives of Gerontology and Geriatrics*, *89*, 104049. doi: 10.1016/j.archger.2020.104049.

Select Conference Presentations

1. **Huang, Y.**, Luo, J., Molenaar, D., & Jeon, M. (2025, April) *Capturing item preknowledge with a response time latent space model framework*. Talk presented at the Annual Meeting of the National Council on Measurement in Education, Denver, CO.
2. **Huang, Y.**, Wang, S., Pan, Y., Lu, X., & Chen, P. (2024, April) *A motivation-based cognitive diagnostic model for insufficient responses detection*. Talk presented at the Annual Meeting of the National Council on Measurement in Education, Philadelphia, PA.
3. Yuan, L., **Huang, Y.**, & Chen, P. (2023, July). *Online calibration for P-MCAT: A neural network based approach*. Talk presented at the International Meeting of the Psychometric Society, College Park, Maryland.
4. **Huang, Y.**, Zhang, T., & Chen, P. (2022, July). *Exploring the structure of speed in cognitive diagnostic models*. Poster presented at the International Meeting of the Psychometric Society, Bologna, Italy.
5. Zhang, T., **Huang, Y.**, & Xin, T. (2022, July). *An analysis of adaptive learning recommendation based on reinforcement learning*. Poster presented at the International Meeting of the Psychometric Society, Bologna, Italy.
6. **Huang, Y.**, Ren, H., & Chen, P. (2022, April). *New item selection designs for computerized classification test*. Flash Talk presented at the Annual Meeting of the National Council on Measurement in Education, San Diego, CA, Virtual Meeting.

Select Funded Grants

2021 Independent Project (Grant No. BJZK-2020A2-20011). **PI**
Item Selection Methods for Computerized Classification Testing 2021 – 2022
Funded by Collaborative Innovation Center of Assessment for Basic Education Quality $1,000

2020 National Natural Science Foundation of China (Grant No. 32071092). **Participant**
Online Calibration in Computerized Adaptive Testing: New Challenges and Solutions 2021 – 2024
Funded by National Natural Science Foundation of China $90,000

2019 Extracurricular Research Project (Grant No. 19XLGA01). **PI**
High Quality of Teacher-Student Relationships Promotes Mathematics Achievement of Junior High School Students: The Multiple Mediation Effect 2019 – 2020
Funded by South China Normal University $500

2018 Special Funds for Seed Breeding Program (Grant No. 18XLGA07). **Co-PI**
Second Language Learning through Mobile Devices: The Effect of Perceived Usefulness, Perceived Ease of Use, and Flow Experience 2018 – 2019
Funded by South China Normal University $500

Research Experience

Jeon lab, Los Angeles, United States
Graduate Student Researcher Dec. 2023 – Present
Prof. Minjeong Jeon's lab centers on developing, applying, and estimating various latent variable models for studying measurement and growth

- developing cultural relevant reading assessment for visually impaired children
- analyzing the NIH-supported annual mHealth Training project with TERGMs
- developing longitudinal latent process model with multiple learning targets

National Assessment Center for Key Technologies, Beijing, China
Research Assistant Sep. 2020 – Jun. 2023
Prof. Ping Chen's lab focuses on building and developing more effective and efficient algorithms for parameter estimation and adaptive testing

- Designed and conducted studies introducing new item selection methods, online calibration designs, and a motivation-based cognitive diagnosis model
- Helped manage the pretesting of items for the National Assessment

National Assessment Guangdong Sub-Center for Item Bank Construction, Guangzhou, China
Research Assistant Sep. 2017 – Jun. 2020
Prof. Minqiang Zhang's lab studies the validity of new Verbal and Mathematics item formats, the score reports system, and test equating in China's College Entrance Examination

- Contributed to the Test Quality Analysis Report (Math) of China's College Entrance Exam
- Assisted with the implementation of the Guangzhou Sunshine education evaluation project and helped write the survey report
- Designed a study on how students' mathematics performance is affected by the quality of the teacher-student relationship, analyzed data from 3,997 students, and led a funded grant

Guangdong Key Laboratory of Mental Health and Cognitive Science, Guangzhou, China
Data Analysis Research Assistant Oct. 2018 – May. 2019
Prof. Aitao Lu's lab studies the mechanism of language acquisition and how human brains execute language processing

- Designed and conducted studies investigating app-based second-language learning, especially for Chinese and English learners
- Collected 786 questionnaires and analyzed the data with latent profile analysis, hierarchical regression, structural equation modeling, and network analysis

Teaching Experience

UCLA, California, United States. *Teaching Assistant*
EDUC 230C: Linear Statistical Models: Analysis of Designed Experiments Mar. 2025 – Jun. 2025
EDUC 230B: Linear Statistical Models: Multiple Regression Analysis Jan. 2025 – Mar. 2025
EDUC 230A: Introduction to Research Design and Statistics Sep. 2024 – Dec. 2024
EDUC 151: Quantitative Research in Education Sep. 2024 – Dec. 2024

Beijing Normal University, Beijing, China. *Teaching Assistant*
Adaptive testing and diagnostic adaptive assessment (graduate course) Sep. 2021 – Jan. 2022
Think and act like a psychometrician (undergraduate course) Feb. 2021 – Jun. 2021

Skills

- **Analytical:** Proficient in R, M*plus*, SPSS, HLM, latentGOLD; Capable of Python and Conquest
- **Design:** Photoshop, HTML, LaTeX, Unity

Select Honors and Awards

Graduate Dean's Scholar Award, UCLA 2024, 2025
Gordon & Olga Smith Scholarship, UCLA 2023
Outstanding Graduates of Beijing, Beijing Municipal Education Commission 2023
Outstanding Graduates, Beijing Normal University 2023
National Scholarship, Minister of Education of China 2022
First-class Academic Scholarship, Beijing Normal University 2021
Second-class Comprehensive Scholarship, South China Normal University 2017 - 2018

Administrative and Other Experience

Team Leader & Psychology Teacher, Shenzhen Hongling Middle School 2019
Organized mental health assessment for 696 students, taught 25 courses, and provided psycholog-

ical counselling to 36 students (Awarded as Excellent Team Leader)

Vice Chairman, The Business Pioneering Camp 2018 – 2019
Planned and organized business competitions, trained new members, and managed the budget (Awarded as Outstanding Student Leader)

Reviewer Experience:
Current Psychology (peer-reviewed journal); National Council on Measurement in Education (NCME) Annual Meeting 2025; American Educational Research Association (AERA) Annual Meeting 2024, 2025, 2026

Volunteer Experience:
Devoted to supporting education at the Wanzi Primary School in Meizhou, an undeveloped county in Southeast China (Awarded as Advanced Individual) [Program Web] Aug. 2017